

MATH 350-2 Advanced Calculus

W.R. Casper

Department of Mathematics
California State University Fullerton

October 1, 2025

Outline

- 1 Real Analysis Lecture 11
 - More on metric spaces
 - Metric subspaces
 - Limits of sequences

Outline

- 1 Real Analysis Lecture 11
 - More on metric spaces
 - Metric subspaces
 - Limits of sequences

Open sets

Definition

Open sets

Definition

Let (M, d) be a metric space and $A \subseteq M$.

Open sets

Definition

Let (M, d) be a metric space and $A \subseteq M$.

A point $x \in A$ is called an **interior point** if there exists $r > 0$ such that $B_M(x; r) \subseteq A$.

Open sets

Definition

Let (M, d) be a metric space and $A \subseteq M$.

A point $x \in A$ is called an **interior point** if there exists $r > 0$ such that $B_M(x; r) \subseteq A$.

The set $\text{int}(A)$ of interior points of A is called the **interior** of A .

Open sets

Definition

Let (M, d) be a metric space and $A \subseteq M$.

A point $x \in A$ is called an **interior point** if there exists $r > 0$ such that $B_M(x; r) \subseteq A$.

The set $\text{int}(A)$ of interior points of A is called the **interior** of A .

The set A is **open** if every point in A is an interior point, or equivalently $\text{int}(A) = A$.

Challenge!

Problem

Consider the metric space $(\mathbb{R}, d_{\text{disc}})$ where d_{disc} is the discrete metric.

What sets are open sets?

Closed sets

Definition

Closed sets

Definition

Let (M, d) be a metric space and $A \subseteq M$.

Closed sets

Definition

Let (M, d) be a metric space and $A \subseteq M$.

A set $A \subseteq M$ is closed if the complement $M \setminus A$ is open.

Closed sets

Definition

Let (M, d) be a metric space and $A \subseteq M$.

A set $A \subseteq M$ is closed if the complement $M \setminus A$ is open.

A point $x \in M$ is an **adherent point** if for all $r > 0$ the ball $B_M(x; r)$ contains an element of A .

Closed sets

Definition

Let (M, d) be a metric space and $A \subseteq M$.

A set $A \subseteq M$ is closed if the complement $M \setminus A$ is open.

A point $x \in M$ is an **adherent point** if for all $r > 0$ the ball $B_M(x; r)$ contains an element of A .

A point $x \in M$ is an **accumulation point** if for all $r > 0$ the ball $B_M(x; r)$ contains an element of A different from x .

Closed sets

Definition

Let (M, d) be a metric space and $A \subseteq M$.

A set $A \subseteq M$ is closed if the complement $M \setminus A$ is open.

A point $x \in M$ is an **adherent point** if for all $r > 0$ the ball $B_M(x; r)$ contains an element of A .

A point $x \in M$ is an **accumulation point** if for all $r > 0$ the ball $B_M(x; r)$ contains an element of A different from x .

The set $\bar{A}$ of all adherent points of A is called the **closure** of A .

Challenge!

Problem

Consider the metric space $(\mathbb{R}, d_{\text{disc}})$ where d_{disc} is the discrete metric.

Which sets are closed?

Challenge!

Problem

Prove the following (Apostol Theorem 3.36). If (M, d) is a metric space, $U \subseteq M$ is open, and $C \subseteq M$ is closed, then $U \setminus C$ is open and $C \setminus U$ is closed.

Characterizing closed sets

Theorem (Apostol Theorem 3.37)

Let (M, d) be a metric space and $A \subseteq S$. Then the following are equivalent:

Characterizing closed sets

Theorem (Apostol Theorem 3.37)

Let (M, d) be a metric space and $A \subseteq S$. Then the following are equivalent:

- (a) *A is closed*

Characterizing closed sets

Theorem (Apostol Theorem 3.37)

Let (M, d) be a metric space and $A \subseteq S$. Then the following are equivalent:

- (a) A is closed*
- (b) A contains all of its adherent points*

Characterizing closed sets

Theorem (Apostol Theorem 3.37)

Let (M, d) be a metric space and $A \subseteq S$. Then the following are equivalent:

- (a) A is closed*
- (b) A contains all of its adherent points*
- (c) A contains all of its accumulation points*

Characterizing closed sets

Theorem (Apostol Theorem 3.37)

Let (M, d) be a metric space and $A \subseteq S$. Then the following are equivalent:

- (a) *A is closed*
- (b) *A contains all of its adherent points*
- (c) *A contains all of its accumulation points*
- (d) *$A = \bar{A}$*

Unions and intersections

Unions and intersections

Theorem

Let (M, d) be a metric space and $\{U_i : i \in I\}$ be a family of open sets in M . The union $\bigcup_{i \in I} U_i$ is open.

Unions and intersections

Theorem

Let (M, d) be a metric space and $\{U_i : i \in I\}$ be a family of open sets in M . The union $\bigcup_{i \in I} U_i$ is open.

Proof.

Identical to the proof for $\mathbb{R}^n$. □

Unions and intersections

Theorem

Let (M, d) be a metric space and $\{U_i : i \in I\}$ be a family of open sets in M . The union $\bigcup_{i \in I} U_i$ is open.

Proof.

Identical to the proof for $\mathbb{R}^n$. □

Theorem

Let (M, d) be a metric space and $U_1, U_2, \dots, U_n$ be open sets in M . The intersection $\bigcap_{i=1}^n U_i$ is open.

Unions and intersections

Theorem

Let (M, d) be a metric space and $\{U_i : i \in I\}$ be a family of open sets in M . The union $\bigcup_{i \in I} U_i$ is open.

Proof.

Identical to the proof for $\mathbb{R}^n$. □

Theorem

Let (M, d) be a metric space and $U_1, U_2, \dots, U_n$ be open sets in M . The intersection $\bigcap_{i=1}^n U_i$ is open.

Proof.

Identical to the proof for $\mathbb{R}^n$. □

Unions and intersections

Unions and intersections

Theorem

Let (M, d) be a metric space and $\{C_i : i \in I\}$ be a family of closed sets in M . The intersection $\bigcap_{i \in I} C_i$ is closed.

Unions and intersections

Theorem

Let (M, d) be a metric space and $\{C_i : i \in I\}$ be a family of closed sets in M . The intersection $\bigcap_{i \in I} C_i$ is closed.

Proof.

Apply De Morgan's Laws.



Unions and intersections

Theorem

Let (M, d) be a metric space and $\{C_i : i \in I\}$ be a family of closed sets in M . The intersection $\bigcap_{i \in I} C_i$ is closed.

Proof.

Apply De Morgan's Laws.



Theorem

Let (M, d) be a metric space and $C_1, C_2, \dots, C_n$ be closed sets in M . The intersection $\bigcap_{i=1}^n C_i$ is closed.

Unions and intersections

Theorem

Let (M, d) be a metric space and $\{C_i : i \in I\}$ be a family of closed sets in M . The intersection $\bigcap_{i \in I} C_i$ is closed.

Proof.

Apply De Morgan's Laws. □

Theorem

Let (M, d) be a metric space and $C_1, C_2, \dots, C_n$ be closed sets in M . The intersection $\bigcup_{i=1}^n C_i$ is closed.

Proof.

Apply De Morgan's Laws. □

Outline

- 1 Real Analysis Lecture 11
 - More on metric spaces
 - **Metric subspaces**
 - Limits of sequences

Metric subspace

Definition

Let (M, d) be a metric space.

Metric subspace

Definition

Let (M, d) be a metric space.

If $S \subseteq M$ is a subset, then $d : M \times M \rightarrow \mathbb{R}$ restricts to a metric $d : S \times S \rightarrow \mathbb{R}$, called the **relative metric**.

Metric subspace

Definition

Let (M, d) be a metric space.

If $S \subseteq M$ is a subset, then $d : M \times M \rightarrow \mathbb{R}$ restricts to a metric $d : S \times S \rightarrow \mathbb{R}$, called the **relative metric**.

The space (S, d) with the relative metric is called a **metric subspace**.

Metric subspace

Definition

Let (M, d) be a metric space.

If $S \subseteq M$ is a subset, then $d : M \times M \rightarrow \mathbb{R}$ restricts to a metric $d : S \times S \rightarrow \mathbb{R}$, called the **relative metric**.

The space (S, d) with the relative metric is called a **metric subspace**.

Open balls in S :

$$B_S(x; r) = B_M(x; r) \cap S.$$

Example

Let $M = \mathbb{R}$ and d be the Euclidean metric.

Example

Let $M = \mathbb{R}$ and d be the Euclidean metric.

Consider the metric subspace (S, d) for $S = [-1, 1) \cup \{2\}$.

Example

Let $M = \mathbb{R}$ and d be the Euclidean metric.

Consider the metric subspace (S, d) for $S = [-1, 1) \cup \{2\}$.

Open balls:

Example

Let $M = \mathbb{R}$ and d be the Euclidean metric.

Consider the metric subspace (S, d) for $S = [-1, 1) \cup \{2\}$.

Open balls:

$$\{2\}$$

Example

Let $M = \mathbb{R}$ and d be the Euclidean metric.

Consider the metric subspace (S, d) for $S = [-1, 1) \cup \{2\}$.

Open balls:

$$\{2\}$$

$$[-1, s), \quad -1 < s < 1$$

Example

Let $M = \mathbb{R}$ and d be the Euclidean metric.

Consider the metric subspace (S, d) for $S = [-1, 1) \cup \{2\}$.

Open balls:

$$\{2\}$$

$$[-1, s), \quad -1 < s < 1$$

$$(r, s), \quad -1 < r < s < 1$$

Example

Let $M = \mathbb{R}$ and d be the Euclidean metric.

Consider the metric subspace (S, d) for $S = [-1, 1) \cup \{2\}$.

Open balls:

$$\{2\}$$

$$[-1, s), \quad -1 < s < 1$$

$$(r, s), \quad -1 < r < s < 1$$

$$[-1, s) \cup \{2\}, \quad -1 < s < 1$$

Example

Let $M = \mathbb{R}$ and d be the Euclidean metric.

Consider the metric subspace (S, d) for $S = [-1, 1) \cup \{2\}$.

Open balls:

$$\{2\}$$

$$[-1, s), \quad -1 < s < 1$$

$$(r, s), \quad -1 < r < s < 1$$

$$[-1, s) \cup \{2\}, \quad -1 < s < 1$$

$$(r, s) \cup \{2\}, \quad -1 < r < s < 1$$

Open sets in subspaces

Theorem

Let (M, d) be a metric space and let (S, d) be a subspace. Then V is an open subset of S if and only if there exists an open subset U of M with $V = U \cap S$.

Open sets in subspaces

Theorem

Let (M, d) be a metric space and let (S, d) be a subspace. Then V is an open subset of S if and only if there exists an open subset U of M with $V = U \cap S$.

Proof.

Open sets in subspaces

Theorem

Let (M, d) be a metric space and let (S, d) be a subspace. Then V is an open subset of S if and only if there exists an open subset U of M with $V = U \cap S$.

Proof.

$\implies$: Assume V is an open subset of S .

Open sets in subspaces

Theorem

Let (M, d) be a metric space and let (S, d) be a subspace. Then V is an open subset of S if and only if there exists an open subset U of M with $V = U \cap S$.

Proof.

$\implies$: Assume V is an open subset of S .

Then for all $x \in V$, there exists $r_x > 0$ such that $B_S(x; r_x) \subseteq V$.

Open sets in subspaces

Theorem

Let (M, d) be a metric space and let (S, d) be a subspace. Then V is an open subset of S if and only if there exists an open subset U of M with $V = U \cap S$.

Proof.

$\implies$: Assume V is an open subset of S .

Then for all $x \in V$, there exists $r_x > 0$ such that $B_S(x; r_x) \subseteq V$.

This says $B_M(x; r_x) \cap S \subseteq V$.

Open sets in subspaces

Theorem

Let (M, d) be a metric space and let (S, d) be a subspace. Then V is an open subset of S if and only if there exists an open subset U of M with $V = U \cap S$.

Proof.

$\Rightarrow$: Assume V is an open subset of S .

Then for all $x \in V$, there exists $r_x > 0$ such that $B_S(x; r_x) \subseteq V$.

This says $B_M(x; r_x) \cap S \subseteq V$.

Define

$$U = \bigcup_{x \in V} B_M(x; r_x).$$

Open sets in subspaces

Theorem

Let (M, d) be a metric space and let (S, d) be a subspace. Then V is an open subset of S if and only if there exists an open subset U of M with $V = U \cap S$.

Proof.

$\implies$: Assume V is an open subset of S .

Then for all $x \in V$, there exists $r_x > 0$ such that $B_S(x; r_x) \subseteq V$.

This says $B_M(x; r_x) \cap S \subseteq V$.

Define

$$U = \bigcup_{x \in V} B_M(x; r_x).$$

This is a union of open sets, and is therefore open.

Open sets in subspaces

Theorem

Let (M, d) be a metric space and let (S, d) be a subspace. Then V is an open subset of S if and only if there exists an open subset U of M with $V = U \cap S$.

Proof.

$\implies$: Assume V is an open subset of S .

Then for all $x \in V$, there exists $r_x > 0$ such that $B_S(x; r_x) \subseteq V$.

This says $B_M(x; r_x) \cap S \subseteq V$.

Define

$$U = \bigcup_{x \in V} B_M(x; r_x).$$

This is a union of open sets, and is therefore open.

Furthermore $U \cap S = V$.



Open sets in subspaces

Theorem

Let (M, d) be a metric space and let (S, d) be a subspace. Then V is an open subset of S if and only if there exists an open subset U of M with $V = U \cap S$.

Open sets in subspaces

Theorem

Let (M, d) be a metric space and let (S, d) be a subspace. Then V is an open subset of S if and only if there exists an open subset U of M with $V = U \cap S$.

Proof.

Open sets in subspaces

Theorem

Let (M, d) be a metric space and let (S, d) be a subspace. Then V is an open subset of S if and only if there exists an open subset U of M with $V = U \cap S$.

Proof.

$\Leftarrow$: Assume U is an open subset of M .

Open sets in subspaces

Theorem

Let (M, d) be a metric space and let (S, d) be a subspace. Then V is an open subset of S if and only if there exists an open subset U of M with $V = U \cap S$.

Proof.

$\Leftarrow$: Assume U is an open subset of M .

We need to show that $V := U \cap S$ is open.

Open sets in subspaces

Theorem

Let (M, d) be a metric space and let (S, d) be a subspace. Then V is an open subset of S if and only if there exists an open subset U of M with $V = U \cap S$.

Proof.

$\Leftarrow$: Assume U is an open subset of M .

We need to show that $V := U \cap S$ is open.

If $x \in V$, then $x \in U$ and there exists $r > 0$ such that $B_M(x; r) \subseteq U$.

Open sets in subspaces

Theorem

Let (M, d) be a metric space and let (S, d) be a subspace. Then V is an open subset of S if and only if there exists an open subset U of M with $V = U \cap S$.

Proof.

$\Leftarrow$: Assume U is an open subset of M .

We need to show that $V := U \cap S$ is open.

If $x \in V$, then $x \in U$ and there exists $r > 0$ such that

$B_M(x; r) \subseteq U$.

It follows that $B_S(x, r) = B_M(x, r) \cap S \subseteq U \cap S = V$.

Open sets in subspaces

Theorem

Let (M, d) be a metric space and let (S, d) be a subspace. Then V is an open subset of S if and only if there exists an open subset U of M with $V = U \cap S$.

Proof.

$\Leftarrow$: Assume U is an open subset of M .

We need to show that $V := U \cap S$ is open.

If $x \in V$, then $x \in U$ and there exists $r > 0$ such that

$B_M(x; r) \subseteq U$.

It follows that $B_S(x, r) = B_M(x, r) \cap S \subseteq U \cap S = V$.

This shows that x is an interior point of S in the metric subspace (S, d) .

Open sets in subspaces

Theorem

Let (M, d) be a metric space and let (S, d) be a subspace. Then V is an open subset of S if and only if there exists an open subset U of M with $V = U \cap S$.

Proof.

$\Leftarrow$: Assume U is an open subset of M .

We need to show that $V := U \cap S$ is open.

If $x \in V$, then $x \in U$ and there exists $r > 0$ such that $B_M(x; r) \subseteq U$.

It follows that $B_S(x, r) = B_M(x, r) \cap S \subseteq U \cap S = V$.

This shows that x is an interior point of S in the metric subspace (S, d) .

Since $x \in V$ was arbitrary, this shows that V is open in S . □

Closed sets in subspaces

Theorem

Let (M, d) be a metric space and let (S, d) be a subspace. Then B is a closed subset of S if and only if there exists a closed subset A of M with $B = A \cap S$.

Closed sets in subspaces

Theorem

Let (M, d) be a metric space and let (S, d) be a subspace. Then B is a closed subset of S if and only if there exists a closed subset A of M with $B = A \cap S$.

Proof.

Closed sets in subspaces

Theorem

Let (M, d) be a metric space and let (S, d) be a subspace. Then B is a closed subset of S if and only if there exists a closed subset A of M with $B = A \cap S$.

Proof.

Let $B \subseteq S$.

Closed sets in subspaces

Theorem

Let (M, d) be a metric space and let (S, d) be a subspace. Then B is a closed subset of S if and only if there exists a closed subset A of M with $B = A \cap S$.

Proof.

Let $B \subseteq S$.

B is closed in S if and only if $S \setminus B$ is open in B

Closed sets in subspaces

Theorem

Let (M, d) be a metric space and let (S, d) be a subspace. Then B is a closed subset of S if and only if there exists a closed subset A of M with $B = A \cap S$.

Proof.

Let $B \subseteq S$.

B is closed in S if and only if $S \setminus B$ is open in B

This is true if and only if there exists an open subset U of M with $S \setminus B = U \cap S$.

Closed sets in subspaces

Theorem

Let (M, d) be a metric space and let (S, d) be a subspace. Then B is a closed subset of S if and only if there exists a closed subset A of M with $B = A \cap S$.

Proof.

Let $B \subseteq S$.

B is closed in S if and only if $S \setminus B$ is open in B

This is true if and only if there exists an open subset U of M with $S \setminus B = U \cap S$.

This is true if and only if there exists an open subset U of M with $B = (M \setminus U) \cap S$.

Closed sets in subspaces

Theorem

Let (M, d) be a metric space and let (S, d) be a subspace. Then B is a closed subset of S if and only if there exists a closed subset A of M with $B = A \cap S$.

Proof.

Let $B \subseteq S$.

B is closed in S if and only if $S \setminus B$ is open in B

This is true if and only if there exists an open subset U of M with $S \setminus B = U \cap S$.

This is true if and only if there exists an open subset U of M with $B = (M \setminus U) \cap S$.

this is true if and only if there exists a closed subset A of M with $B = A \cap S$. □

Example

Consider the metric space $(\mathbb{R}, d)$ with $d = d_{\text{eucl}}$ the Euclidean metric, and the metric subspace (S, d) for $S = [0, 1]$.

Example

Consider the metric space $(\mathbb{R}, d)$ with $d = d_{\text{eucl}}$ the Euclidean metric, and the metric subspace (S, d) for $S = [0, 1]$.

- $[0, 1]$ is open in S and closed in S , but it is not open in M

Example

Consider the metric space $(\mathbb{R}, d)$ with $d = d_{\text{eucl}}$ the Euclidean metric, and the metric subspace (S, d) for $S = [0, 1]$.

- $[0, 1]$ is open in S and closed in S , but it is not open in M
- $[0, 1/2)$ is open in S , but not in M

Example

Consider the metric space $(\mathbb{R}, d)$ with $d = d_{\text{eucl}}$ the Euclidean metric, and the metric subspace (S, d) for $S = [0, 1]$.

- $[0, 1]$ is open in S and closed in S , but it is not open in M
- $[0, 1/2)$ is open in S , but not in M
- $(0, 1/2)$ is open in both S and in M

Example

Consider the metric space $(\mathbb{R}, d)$ with $d = d_{\text{eucl}}$ the Euclidean metric, and the metric subspace (S, d) for $S = [0, 1]$.

- $[0, 1]$ is open in S and closed in S , but it is not open in M
- $[0, 1/2)$ is open in S , but not in M
- $(0, 1/2)$ is open in both S and in M
- $(1/2, 1]$ is open in S , but not in M

Outline

- 1 Real Analysis Lecture 11
 - More on metric spaces
 - Metric subspaces
 - Limits of sequences

Definition of a limit

Definition of a limit

Back in Calc. I:

Definition of a limit

Back in Calc. I:

$$\lim_{n \rightarrow \infty} x_n = L \quad \text{means}$$

Definition of a limit

Back in Calc. I:

$$\lim_{n \rightarrow \infty} x_n = L \quad \text{means}$$

that as n gets arbitrarily large, the value of x_n gets arbitrarily close to L .

Definition of a limit

Back in Calc. I:

$$\lim_{n \rightarrow \infty} x_n = L \quad \text{means}$$

that as n gets arbitrarily large, the value of x_n gets arbitrarily close to L .

Formally:

Definition of a limit

Back in Calc. I:

$$\lim_{n \rightarrow \infty} x_n = L \quad \text{means}$$

that as n gets arbitrarily large, the value of x_n gets arbitrarily close to L .

Formally:

$\forall \epsilon > 0$, there exists $N \in \mathbb{Z}_+$ such that $|x_n - L| < \epsilon \quad \forall n \geq N$.

Definition of a limit

Back in Calc. I:

$$\lim_{n \rightarrow \infty} x_n = L \quad \text{means}$$

that as n gets arbitrarily large, the value of x_n gets arbitrarily close to L .

Formally:

$\forall \epsilon > 0$, there exists $N \in \mathbb{Z}_+$ such that $|x_n - L| < \epsilon \forall n \geq N$.

Metric space version:

Definition of a limit

Back in Calc. I:

$$\lim_{n \rightarrow \infty} x_n = L \quad \text{means}$$

that as n gets arbitrarily large, the value of x_n gets arbitrarily close to L .

Formally:

$$\forall \epsilon > 0, \quad \text{there exists } N \in \mathbb{Z}_+ \text{ such that } |x_n - L| < \epsilon \quad \forall n \geq N.$$

Metric space version:

$$\forall \epsilon > 0, \quad \text{there exists } N \in \mathbb{Z}_+ \text{ such that } d(x_n, L) < \epsilon \quad \forall n \geq N.$$

Definition of a limit

Back in Calc. I:

$$\lim_{n \rightarrow \infty} x_n = L \quad \text{means}$$

that as n gets arbitrarily large, the value of x_n gets arbitrarily close to L .

Formally:

$$\forall \epsilon > 0, \quad \text{there exists } N \in \mathbb{Z}_+ \text{ such that } |x_n - L| < \epsilon \quad \forall n \geq N.$$

Metric space version:

$$\forall \epsilon > 0, \quad \text{there exists } N \in \mathbb{Z}_+ \text{ such that } d(x_n, L) < \epsilon \quad \forall n \geq N.$$

We say $\{x_n\}$ **converges** to L and L is the **limit of the sequence**.

Example

Example

Consider the metric space (M, d) defined by $M = \mathbb{R}$ with the Euclidean metric d .

Example

Consider the metric space (M, d) defined by $M = \mathbb{R}$ with the Euclidean metric d .

The sequence $\{x_n\}$ defined by $x_n = 1/n$ converges to zero.

Example

Consider the metric space (M, d) defined by $M = \mathbb{R}$ with the Euclidean metric d .

The sequence $\{x_n\}$ defined by $x_n = 1/n$ converges to zero.

Proof.

Example

Consider the metric space (M, d) defined by $M = \mathbb{R}$ with the Euclidean metric d .

The sequence $\{x_n\}$ defined by $x_n = 1/n$ converges to zero.

Proof.

Let $\epsilon > 0$.

Example

Consider the metric space (M, d) defined by $M = \mathbb{R}$ with the Euclidean metric d .

The sequence $\{x_n\}$ defined by $x_n = 1/n$ converges to zero.

Proof.

Let $\epsilon > 0$.

Choose $N \in \mathbb{Z}_+$ with $N > 1/\epsilon$.

Example

Consider the metric space (M, d) defined by $M = \mathbb{R}$ with the Euclidean metric d .

The sequence $\{x_n\}$ defined by $x_n = 1/n$ converges to zero.

Proof.

Let $\epsilon > 0$.

Choose $N \in \mathbb{Z}_+$ with $N > 1/\epsilon$.

Then $n > N$ implies

Example

Consider the metric space (M, d) defined by $M = \mathbb{R}$ with the Euclidean metric d .

The sequence $\{x_n\}$ defined by $x_n = 1/n$ converges to zero.

Proof.

Let $\epsilon > 0$.

Choose $N \in \mathbb{Z}_+$ with $N > 1/\epsilon$.

Then $n > N$ implies

$$d(x_n, 0) = |1/n - 0| = 1/n < 1/N < \epsilon.$$

Example

Consider the metric space (M, d) defined by $M = \mathbb{R}$ with the Euclidean metric d .

The sequence $\{x_n\}$ defined by $x_n = 1/n$ converges to zero.

Proof.

Let $\epsilon > 0$.

Choose $N \in \mathbb{Z}_+$ with $N > 1/\epsilon$.

Then $n > N$ implies

$$d(x_n, 0) = |1/n - 0| = 1/n < 1/N < \epsilon.$$

Since $\epsilon > 0$ was arbitrary, this proves $\{x_n\}$ converges to 0.

Example

Consider the metric space (M, d) defined by $M = \mathbb{R}$ with the Euclidean metric d .

The sequence $\{x_n\}$ defined by $x_n = 1/n$ converges to zero.

Proof.

Let $\epsilon > 0$.

Choose $N \in \mathbb{Z}_+$ with $N > 1/\epsilon$.

Then $n > N$ implies

$$d(x_n, 0) = |1/n - 0| = 1/n < 1/N < \epsilon.$$

Since $\epsilon > 0$ was arbitrary, this proves $\{x_n\}$ converges to 0. $\square$

Example

Consider the metric space (M, d) defined by $M = \mathbb{R}$ with the Euclidean metric d .

Example

Consider the metric space (M, d) defined by $M = \mathbb{R}$ with the Euclidean metric d .

The sequence $\{x_n\}$ defined by $x_n = (n - 1)/n$ converges to 1.

Example

Consider the metric space (M, d) defined by $M = \mathbb{R}$ with the Euclidean metric d .

The sequence $\{x_n\}$ defined by $x_n = (n - 1)/n$ converges to 1.

Proof.

Example

Consider the metric space (M, d) defined by $M = \mathbb{R}$ with the Euclidean metric d .

The sequence $\{x_n\}$ defined by $x_n = (n - 1)/n$ converges to 1.

Proof.

Let $\epsilon > 0$.

Example

Consider the metric space (M, d) defined by $M = \mathbb{R}$ with the Euclidean metric d .

The sequence $\{x_n\}$ defined by $x_n = (n - 1)/n$ converges to 1.

Proof.

Let $\epsilon > 0$.

Choose $N \in \mathbb{Z}_+$ with $N > 1/\epsilon$.

Example

Consider the metric space (M, d) defined by $M = \mathbb{R}$ with the Euclidean metric d .

The sequence $\{x_n\}$ defined by $x_n = (n - 1)/n$ converges to 1.

Proof.

Let $\epsilon > 0$.

Choose $N \in \mathbb{Z}_+$ with $N > 1/\epsilon$.

Then $n > N$ implies

$$d(x_n, 1) = |(n - 1)/n - 1| = 1/n < 1/N < \epsilon.$$

Example

Consider the metric space (M, d) defined by $M = \mathbb{R}$ with the Euclidean metric d .

The sequence $\{x_n\}$ defined by $x_n = (n - 1)/n$ converges to 1.

Proof.

Let $\epsilon > 0$.

Choose $N \in \mathbb{Z}_+$ with $N > 1/\epsilon$.

Then $n > N$ implies

$$d(x_n, 1) = |(n - 1)/n - 1| = 1/n < 1/N < \epsilon.$$

Since $\epsilon > 0$ was arbitrary, this proves $\{x_n\}$ converges to 1. □

Example

Consider the metric space (M, d) defined by $M = \mathbb{R}$ with the **discrete** metric d .

Example

Consider the metric space (M, d) defined by $M = \mathbb{R}$ with the **discrete** metric d .

The sequence $\{x_n\}$ defined by $x_n = 1/n$ does not converge.

Example

Consider the metric space (M, d) defined by $M = \mathbb{R}$ with the **discrete** metric d .

The sequence $\{x_n\}$ defined by $x_n = 1/n$ does not converge.

Proof.

Example

Consider the metric space (M, d) defined by $M = \mathbb{R}$ with the **discrete** metric d .

The sequence $\{x_n\}$ defined by $x_n = 1/n$ does not converge.

Proof.

Suppose that it converges to a number L .

Example

Consider the metric space (M, d) defined by $M = \mathbb{R}$ with the **discrete** metric d .

The sequence $\{x_n\}$ defined by $x_n = 1/n$ does not converge.

Proof.

Suppose that it converges to a number L .

Then for all $\epsilon > 0$, there must exist an $N \in \mathbb{Z}_+$ such that $d(x_n, L) < \epsilon$ for all $N \geq n$.

Example

Consider the metric space (M, d) defined by $M = \mathbb{R}$ with the **discrete** metric d .

The sequence $\{x_n\}$ defined by $x_n = 1/n$ does not converge.

Proof.

Suppose that it converges to a number L .

Then for all $\epsilon > 0$, there must exist an $N \in \mathbb{Z}_+$ such that $d(x_n, L) < \epsilon$ for all $N \geq n$.

Take $\epsilon = 1$ and choose $N \in \mathbb{Z}_+$ such that $d(x_n, L) < 1$ for all $N \geq n$.

Example

Consider the metric space (M, d) defined by $M = \mathbb{R}$ with the **discrete** metric d .

The sequence $\{x_n\}$ defined by $x_n = 1/n$ does not converge.

Proof.

Suppose that it converges to a number L .

Then for all $\epsilon > 0$, there must exist an $N \in \mathbb{Z}_+$ such that $d(x_n, L) < \epsilon$ for all $N \geq n$.

Take $\epsilon = 1$ and choose $N \in \mathbb{Z}_+$ such that $d(x_n, L) < 1$ for all $N \geq n$.

With the discrete metric $d(x_n, L) < 1$ implies $x_n = L$.

Example

Consider the metric space (M, d) defined by $M = \mathbb{R}$ with the **discrete** metric d .

The sequence $\{x_n\}$ defined by $x_n = 1/n$ does not converge.

Proof.

Suppose that it converges to a number L .

Then for all $\epsilon > 0$, there must exist an $N \in \mathbb{Z}_+$ such that $d(x_n, L) < \epsilon$ for all $N \geq n$.

Take $\epsilon = 1$ and choose $N \in \mathbb{Z}_+$ such that $d(x_n, L) < 1$ for all $N \geq n$.

With the discrete metric $d(x_n, L) < 1$ implies $x_n = L$.

This means $x_n = 1/n = L$ for all $n \geq N$, which is a contradiction! □

Uniqueness of limits

Theorem (Uniqueness of limits)

Uniqueness of limits

Theorem (Uniqueness of limits)

Let (M, d) be a metric space.

Uniqueness of limits

Theorem (Uniqueness of limits)

Let (M, d) be a metric space.

If a sequence $\{x_n\}$ in M converges to L and converges to M , then $L = M$.

Uniqueness of limits

Theorem (Uniqueness of limits)

Let (M, d) be a metric space.

If a sequence $\{x_n\}$ in M converges to L and converges to M , then $L = M$.

Proof.

Uniqueness of limits

Theorem (Uniqueness of limits)

Let (M, d) be a metric space.

If a sequence $\{x_n\}$ in M converges to L and converges to M , then $L = M$.

Proof.

Suppose that $\{x_n\}$ converges to L and M .

Uniqueness of limits

Theorem (Uniqueness of limits)

Let (M, d) be a metric space.

If a sequence $\{x_n\}$ in M converges to L and converges to M , then $L = M$.

Proof.

Suppose that $\{x_n\}$ converges to L and M .

Assume further that $L \neq M$ and let $\epsilon = d(L, M)/2$.

Uniqueness of limits

Theorem (Uniqueness of limits)

Let (M, d) be a metric space.

If a sequence $\{x_n\}$ in M converges to L and converges to M , then $L = M$.

Proof.

Suppose that $\{x_n\}$ converges to L and M .

Assume further that $L - M \neq 0$ and let $\epsilon = d(L, M)/2$.

Then there exists N_1 such that $d(x_n, L) < \epsilon$ for $n > N_1$.

Uniqueness of limits

Theorem (Uniqueness of limits)

Let (M, d) be a metric space.

If a sequence $\{x_n\}$ in M converges to L and converges to M , then $L = M$.

Proof.

Suppose that $\{x_n\}$ converges to L and M .

Assume further that $L \neq M$ and let $\epsilon = d(L, M)/2$.

Then there exists N_1 such that $d(x_n, L) < \epsilon$ for $n > N_1$.

Likewise, there exists N_2 such that $d(x_n, M) < \epsilon$ for $n > N_2$.

Uniqueness of limits

Theorem (Uniqueness of limits)

Let (M, d) be a metric space.

If a sequence $\{x_n\}$ in M converges to L and converges to M , then $L = M$.

Proof.

Suppose that $\{x_n\}$ converges to L and M .

Assume further that $L \neq M$ and let $\epsilon = d(L, M)/2$.

Then there exists N_1 such that $d(x_n, L) < \epsilon$ for $n > N_1$.

Likewise, there exists N_2 such that $d(x_n, M) < \epsilon$ for $n > N_2$.

Then for $n > \max\{N_1, N_2\}$, the triangle inequality says

Uniqueness of limits

Theorem (Uniqueness of limits)

Let (M, d) be a metric space.

If a sequence $\{x_n\}$ in M converges to L and converges to M , then $L = M$.

Proof.

Suppose that $\{x_n\}$ converges to L and M .

Assume further that $L \neq M$ and let $\epsilon = d(L, M)/2$.

Then there exists N_1 such that $d(x_n, L) < \epsilon$ for $n > N_1$.

Likewise, there exists N_2 such that $d(x_n, M) < \epsilon$ for $n > N_2$.

Then for $n > \max\{N_1, N_2\}$, the triangle inequality says

$$d(L, M) \leq d(L, x_n) + d(x_n, M) < 2\epsilon = d(L, M).$$

Uniqueness of limits

Theorem (Uniqueness of limits)

Let (M, d) be a metric space.

If a sequence $\{x_n\}$ in M converges to L and converges to M , then $L = M$.

Proof.

Suppose that $\{x_n\}$ converges to L and M .

Assume further that $L \neq M$ and let $\epsilon = d(L, M)/2$.

Then there exists N_1 such that $d(x_n, L) < \epsilon$ for $n > N_1$.

Likewise, there exists N_2 such that $d(x_n, M) < \epsilon$ for $n > N_2$.

Then for $n > \max\{N_1, N_2\}$, the triangle inequality says

$$d(L, M) \leq d(L, x_n) + d(x_n, M) < 2\epsilon = d(L, M).$$

This is a contradiction.

